

Analytical and numerical modelling of wave propagation in (meta)poro(elastic) layers

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This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 | Modelling of porous media

Porous media is a category of materials which encompasses naturally occurring substances such as rocks, soils and biological tissues, and engineered materials like ceramics, metallic foams, rockwool or certain types of polymers. They are considered biphasic, that is constituted of two distinct domains: a pore network Ω_f filled by a saturating fluid embedded in an elastic matrix Ω_s . A diagram of this microscopic structure is shown in Fig. 1.1. The porosity is defined as the fraction of fluid over the total volume of the material, as $\phi = V(\Omega_f)/V(\Omega)$. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. At microscopic scales, the geometry of the fluid phase thus forms a complex microstructure. Fortunately, at airborne acoustic scales, these materials can often be considered at a meso-/macroscopic level and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that accounts for their complex geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through friction (viscous losses) along the duct/pore network, while thermal exchanges and elastic dissipation take place when the skeleton and the fluid phase are coupled. These losses are thus dependant on the geometry, which parametrized by the porosity, the saturating fluid properties, the pore network size, connectivity and shape, as well as the frame mechanical parameters.

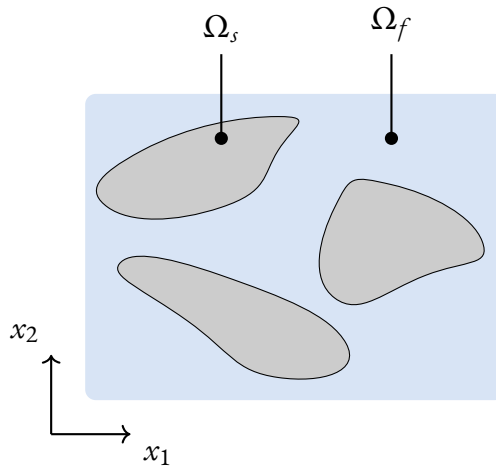


Fig. 1.1: Porous material structure

1.1.1. Rigid-frame approximation

In the so-called rigid-frame approximation, the skeleton of the porous material is considered motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. A harmonic acoustic propagation is assumed with a positive time convention $e^{i\omega t}$ throughout this manuscript (with the exception of Chapter ??). The Helmholtz equation governs the propagation,

$$(\nabla^2 + k_{eq}^2)p(\mathbf{x}) = 0, \quad (1.1)$$

where p is the pressure field and k_{eq} is the wavenumber of the equivalent fluid. In the case where porous or fibrous materials have a porosity close to 1 and the frequency range of interest is limited, the Delany-Bazley model was originally developed in 1970 to propose empirical expressions for the values of the complex wavenumber and characteristic impedances^{1,2} and later improved by Miki³. Although limited in accuracy and validity domain, these models have been widely used since they only required the flow resistivity of the porous material. Later on, a more precise semi-phenomenological model additionally accounting for the viscous and inertial dissipative effects was developed from 1987 to 1991. It is the Johnson-Champoux-Allard (JCA) model which models the complex effective

density $\rho_{eq}(\omega)$ accounts for the frequency dependant viscous losses and effective bulk modulus of the material K_{eq} accounting for thermal exchanges in the layer as,

$$\begin{aligned}\rho_{eq}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - i \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda} \right)^2} \right), \\ K_{eq} &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) / \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \text{Pr} \omega} \sqrt{1 - i \text{Pr} \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1}.\end{aligned}\quad (1.2)$$

These expressions involve a large number of effective parameters for the porous material; α_∞ is the tortuosity, R_f is the flow resistivity, Λ and Λ' are respectively the viscous and thermal characteristic length. The saturating fluid is characterized in this model by its heat capacity ratio κ (commonly also denoted γ), density ρ_f , kinematic viscosity μ_f , pressure P_0 and Prandtl number $\text{Pr} = c_p \mu_f / k$ with c_p is the specific heat and k the thermal conductivity. We can introduce also the Biot frequency $\omega_c = R_f \phi / (\rho_f \alpha_\infty)$ that defines the transition frequency between the inertial and viscous regimes.

1.1.2. Elastic motion of the skeleton: Biot theory

While the equivalent fluid formalism is able to accurately model porous material in specific sets of conditions, that is when the skeleton can be assumed to be motionless, this assumption can quickly become limited as soon as the said skeleton is subjected to an elastic deformation. This is can be the case when the saturating fluid of the PEM has a high density (e.g. water), when the skeleton is coupled to elastic plates or when the acoustic radiation has a wavelength comparable to the pore scale, although these are just examples. In this situation, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory⁴⁻⁷ is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, the various models will be briefly be presented along with the relevant physical quantities. A full derivation would be out of scope and is presented in existing works².

Before presenting the full Biot model, some quantities should be provided additionally to the equivalent fluid parameters. The frame is assumed to undergo elastic deformations, thus the common mechanical parameters used are the solid density of the frame material ρ_s as well as its apparent density $\rho_1 = (1 - \phi) \rho_s$. The bulk moduli are denoted K_b and K_s , respectively for the frame and the material. The shear modulus is also considered complex, accounting for structural losses as $\tilde{N} = N(1 + i\eta)$ with η the loss factor. Symbols written with a tilde denote complex expressions and therefore account for some sort of loss mechanisms. For the fluid, the apparent density is thus $\rho_2 = \phi \rho_f$ and the inertial interaction density defined by $\rho_{12} = -\phi \rho_f (\alpha_\infty - 1)$. A summary on Biot theory is also proposed in Chap. ??) with the tilde symbols omitted for ease of reading.

Formulations for the motion equations

The Biot poroelastic equations from the original 1956 formulation^{4,6,7} depend on the solid displacement vector $\mathbf{u}_s$ and the fluid displacement $\mathbf{u}_f$,

$$\begin{aligned}\nabla \cdot \boldsymbol{\sigma}_s(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{11} \mathbf{u}_s - \omega^2 \tilde{\rho}_{12} \mathbf{u}_f, \\ \nabla \cdot \boldsymbol{\sigma}_f(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{12} \mathbf{u}_s - \omega^2 \tilde{\rho}_{22} \mathbf{u}_f,\end{aligned}\quad (1.3)$$

where $\boldsymbol{\sigma}^s$ is the solid stress tensor, $\boldsymbol{\sigma}^f$ the fluid stress tensor and the apparent densities of the solid-phase is $\tilde{\rho}_{11}$, that of the fluid is $\tilde{\rho}_{22}$ and the coupling apparent density is $\tilde{\rho}_{12}$. They are complex

quantities expression that also account for the contribution of viscous forces and are given as,

$$\tilde{\rho}_{22} = \phi^2 \tilde{\rho}_{eq}, \quad \tilde{\rho}_{12} = \rho_2 - \tilde{\rho}_{22}, \quad \tilde{\rho}_{11} = \rho_1 - \tilde{\rho}_{12}. \quad (1.4)$$

These coupled motion equations are associated to the stress-strain relation,

$$\begin{aligned} \sigma_{s,ij} &= [(\tilde{P} - 2\tilde{N})\nabla \cdot \mathbf{u}_s + \tilde{Q}\nabla \cdot \mathbf{u}_f] \delta_{ij} + 2\tilde{N}\epsilon_{s,ij}, \\ \sigma_{f,ij} &= (\tilde{Q}\nabla \cdot \mathbf{u}_s + \tilde{R}\nabla \cdot \mathbf{u}_f) \delta_{ij} = -\phi p, \end{aligned} \quad (1.5)$$

where $\nabla \cdot \mathbf{u}_s$ and $\nabla \cdot \mathbf{u}^f$ correspond to the dilatation of the solid and fluid phases respectively and ϵ_s , is the solid-phase strain tensor. The Biot coefficients are denoted as $\tilde{P}$ for the solid elastic coefficient, $\tilde{Q}$ for the elastic-fluid coupling coefficient, and $\tilde{R}$ for the fluid-phase coefficient. In the context of a PEM saturated by a light fluid, their expressions are

$$\tilde{P} = K_b + \frac{4}{3}N + (1 - \phi)^2 \frac{\tilde{K}_{eq}}{\phi}, \quad \tilde{Q} = \tilde{K}_{eq}(1 - \phi), \quad \tilde{R} = \phi \tilde{K}_{eq}. \quad (1.6)$$

The full coefficients are generalized forms required when the limit $K_b/K_s \rightarrow 0$ no longer holds, as is the case for water-saturated PEM.

A different formulation, especially convenient for finite-elements discretization is known as the $(\mathbf{u}^s, p)$ formulation. First, new parameters in terms of the Biot elastic coefficients and apparent densities can be defined,

$$\hat{A} = \tilde{P} - 2\tilde{N}_s - \frac{\tilde{Q}^2}{\tilde{R}}, \quad \hat{\rho} = \tilde{\rho}_{11} - \frac{\tilde{\rho}_{12}^2}{\tilde{\rho}_{22}}, \quad \beta = \phi \left(\frac{\tilde{\rho}_{12}}{\tilde{\rho}_{22}} - \frac{\tilde{Q}}{\tilde{R}} \right). \quad (1.7)$$

β is another coupling coefficient, also denoted as $\tilde{\gamma}$ in the literature. Using the second row of Eq. (??) allows for the expression of the fluid displacement as, $\mathbf{u}^f = \phi/(\tilde{\rho}_{22}\omega^2)\nabla p - \tilde{\rho}_{12}/\tilde{\rho}_{22}\mathbf{u}^s$ and using that in Eq. (??) and Eq. (1.5) allow to rewrite the coupled motion equations. Using these, the so-called $(\mathbf{u}_s, p)$ formulation of the Biot theory⁸ is expressed as,

$$\begin{aligned} \nabla \cdot \hat{\sigma}_s + \hat{\rho}\omega^2 \mathbf{u}_s + \beta \nabla p &= 0, \\ \frac{\nabla^2 p}{\rho_{22}\omega^2} - \frac{\beta}{\phi^2} \nabla \cdot \mathbf{u}_s + \frac{1}{R} p &= 0, \end{aligned} \quad (1.8)$$

where $\hat{\sigma}_s$ is the *in vacuo* solid stress tensor,

$$\hat{\sigma}_{s,ij} = \sigma_{s,ij} - \phi \frac{\tilde{Q}}{\tilde{R}} p \delta_{ij} = \hat{A} \nabla \cdot \mathbf{u}_s \delta_{ij} + 2\tilde{N}_s \epsilon_{s,ij}. \quad (1.9)$$

Propagation in unbounded poroelastic media

These two formulations provide equivalent ways of modelling wave propagation in poroelastic media. It appears that PEM supports three waves: a compression φ_p and shear wave ψ_s related to the solid phase as well as an additional compression wave φ_a , related to the fluid phase. In the case of each of these formulations, the fields can be expressed depending on these potentials,

$$\mathbf{u}_s = \nabla \varphi_p + \nabla \varphi_a + \nabla \times \psi_s, \quad \mathbf{u}_f = \mu_p \nabla \varphi_p + \mu_a \nabla \varphi_a + \mu_s \nabla \times \psi_s, \quad (1.10)$$

with μ_p, μ_a, μ_s are coefficients determined by the ratio of solid over fluid displacements ratio. The wavenumbers for each of these waves are, following section 6.5 of Allard and Atalla textbook² and denoted as $\delta_i, i = 1, 2, 3$ in the text,

$$\begin{aligned} k_{p,a} &= \sqrt{\frac{1}{2} \left(\delta_2^2 + \delta_{eq}^2 \pm \delta_t \right)}, \\ k_s &= \omega \sqrt{\frac{\hat{\rho}}{\tilde{N}}}, \end{aligned} \quad (1.11)$$

respectively for the solid compression, fluid compression and shear wave, with,

$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}}}, \quad \delta_1 = \omega \sqrt{\frac{\hat{\rho}}{\hat{P}}}, \quad \delta_2 = \omega \sqrt{\frac{\hat{\rho}_s}{\hat{P}}}, \quad \delta_t = \sqrt{(\delta_2^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2 \delta_1^2}. \quad (1.12)$$

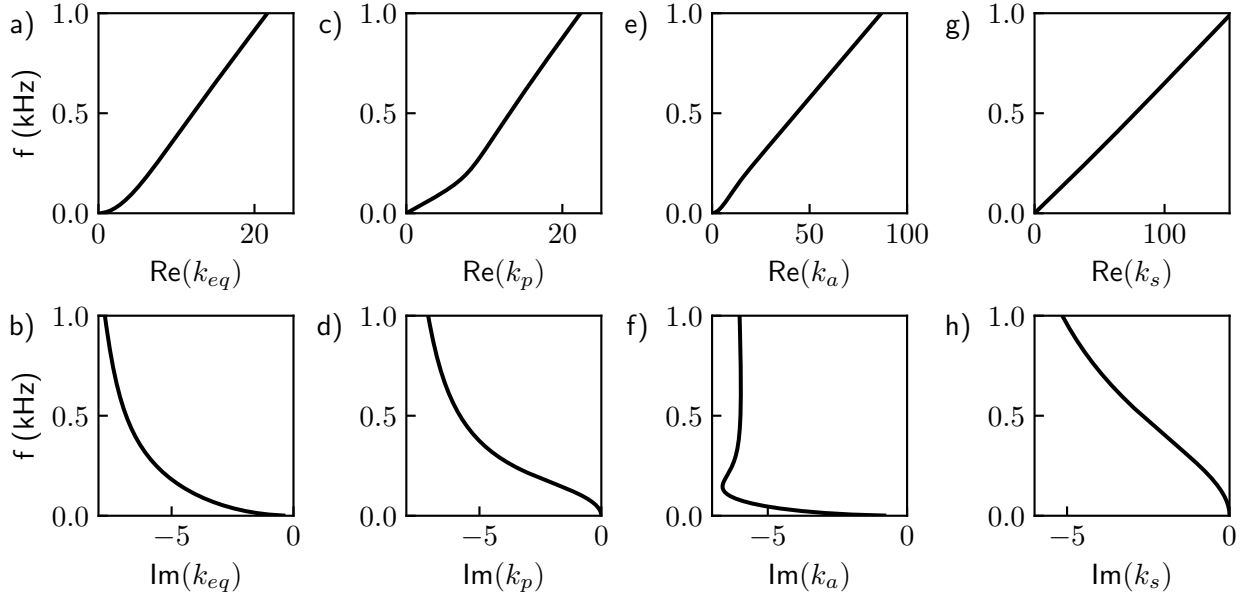


Fig. 1.2: Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.

The propagation characteristics, i.e. wavenumbers of the waves propagating in porous and poroelastic media are illustrated in Figure 1.2. As discussed in the previous sections, the wavenumber for the equivalent fluid as well as the three for the poroelastic layer are complex. Their real part account for the propagation of the phase flow, inversly proportional to the wave velocity and their imaginary part accounts for the attenuation that occurs due to the losses inherent to each of these materials. It appears that at low frequencies, the attenuation is very low for all waves. The velocity of the compression and shear waves (observed on the figures by the slope of the real parts of the wavenumbers) is much higher than the waves in the fluid for the equivalent fluid and the compression fluid phase velocity, as they propagate in a soft solid where the velocity is usually much higher than in a fluid.

1.2 | Wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, impinging at the surface of a structure and is then scattered (reflected and transmitted) from and through each interface.

If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is referred to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they are said to be guided along the layer.

These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often leads to writing complex system that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave by a homogeneous porous layer is investigated along with the various method available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

1.2.1. Analytical plane wave modelling

Modeling acoustic propagation in homogeneous layers by assuming plane waves is the simplest formalism in acoustics. We assume these plane waves such that they satisfy the motion equations, e.g. the ones introduced in eq. (1.1) or eq. (1.3). Introducing a simple problem, that of the scattering of an incident plate by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\mathbf{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1 , there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements.

Assuming a $e^{i\omega t}$ time-dependance, the incident pressure field originated from the upper domain that satisfy the Helmholtz equation takes the form of,

$$p_{0+} = a_+ e^{-ik_1 x_1} e^{-ik_{2,0}(x_2-h)} + a_- e^{-ik_1 x_1} e^{ik_{2,0}(x_2-h)}. \quad (1.13)$$

The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1 x_1} e^{-ik_{2,eq} x_2} + b_- e^{-ik_1 x_1} e^{ik_{2,eq} x_2}, \quad (1.14)$$

and that in the lower air domain is,

$$p_{0-} = c e^{-ik_1 x_1} e^{ik_{2,0} x_2}, \quad (1.15)$$

with wavenumbers such that they hold for the Snell-Descartes law that states

$$k_1^2 + k_{2,0}^2 = k_0^2, \text{ and } k_1^2 + k_{2,eq}^2 = k_{eq}^2. \quad (1.16)$$

The wavenumber should be imposed with $k_1 = k_0 \sin \theta$, $k_2 = -k_0 \cos \theta$, with θ the angle of incidence, that governs the direction of propagation of the incident field. The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -i\omega \rho_i u_2$, $i = (0, eq)$.

Next, the interface conditions describe the relationships between properties at the boundary between two media, including the continuity or vanishing of given fields. If a media is terminated by a surface (e.g. a rigid wall), a constraint should be applied on the fields to represent it, denoted as boundary conditions. Assuming a continuity between the two fluids in the present case with interfaces located at $x_2 = 0$ and $x_2 = h$, we obtain the following conditions,

$$p^t(h) = p^b(h), \quad u_2^t(h) = u_2^b(h) \quad p^l(0) = p^b(0) \quad \frac{l}{2}(0) = u_2^b(0), \quad (1.17)$$

with superscripts b and t denoting respectively the bottom and top interfaces.

The full analytical description of a system constituted from one or several interfaces can be done by writing a linear system built from the interface conditions, which vector of unknowns are the

amplitudes of each of the existing waves in the system, that is $(b_+ \ b_- \ a_+ \ c)^T$. The forcing right-hand side corresponds to the incident wave terms. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients. Imposing the interface conditions leads to writing the following matrix system,

$$\begin{pmatrix} 1 & e^{ik_{2,eq}h} & 0 & -1 \\ \frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}h} & 0 & \frac{k_{2,0}}{\rho_0} \\ e^{ik_{2,eq}h} & 1 & -1 & 0 \\ \frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}h} & -\frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ a_+ \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_- \\ \frac{k_{2,eq}}{\rho_{eq}}a_- \end{pmatrix}. \quad (1.18)$$

From this expression, the system can simply be solved to obtain all scattered amplitudes b_+ , b_- , a_+ and c . In the case of a rigidly-backed layer, the interface at $x_2 = h$ is replaced by a rigid termination. For (equivalent) fluids, a hard wall or Neumann-type boundary condition is translated by a vanishing normal particle displacement field along the boundary, that is $u_2(\Gamma) = 0$. The term depending on c thus vanishes, since p_{0-} does not exist; this reduces the size of the system. In this case, the reflection coefficient R corresponds to the amplitude a_+ .

This method is denoted in this work as the Global Method (GM), since all of the available data is used in the formulation of the problem as the unknowns are all wave amplitudes. As described in Section?? of Chapter ??, a general formalism can be used for this method and any stack of any layer can be described with the GM, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\mathbf{M}(\omega)\mathbf{x} = \mathbf{f}, \quad (1.19)$$

where $\mathbf{M}$ corresponds to the interface conditions involved in the system, $\mathbf{x}$ contain the wave amplitudes and $\mathbf{f}$ is the forcing term, the incident plane wave amplitudes.

In the case of a PEM, the interface conditions are,

$$\sigma_{12}^s = 0, \quad \sigma_{22}^s + \sigma_{22}^f = -p_0, \quad \sigma_{22}^f = -\phi p_0, \quad \phi u_2^f + (1 - \phi)u_2^s = u_2^f. \quad (1.20)$$

and for a rigid-backing,

$$u_1^s(\Gamma) = 0, \quad u_2^s(\Gamma) = 0, \quad u_2^f(\Gamma) = 0. \quad (1.21)$$

⁹ **relire ça.** The system used to obtain the reflection coefficient for the PEM layer is done in a similar fashion as the equivalent fluid and can be read in the appendices of Ref.¹⁰.

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in the case of a symmetric configuration (with air coupling at both interfaces) are represented by the diagrams in Figs 1.3a and b and results in Figs 1.3c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^\circ$ angle. The parameters for each material corresponds to the one used in Table ?? from Chapter ??.

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

1.2.2. Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson¹¹ and Haskell¹² in order to describe elastic wave propagation

1.2. Wave propagation in homogeneous layers

in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics. TMM allows for an easy multiphysics coupling with interface matrices; its core lies in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface, which is possible thanks to the interface conditions. Then the physical field amplitudes stored in the state vector are propagated from one layer to the next. The transfer matrix T relating the pressure and normal displacement fields from the bottom (located at $x_2 = 0$) to the top interface (located at $x_2 = b$) of a fluid layer is thus,¹³

$$\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix} = \underbrace{\begin{pmatrix} \cos(k_{eq}b) & iZ_{eq} \sin(k_{eq}b) \\ iZ_{eq}^{-1} \sin(k_{eq}b) & \cos(k_{eq}b) \end{pmatrix}}_T \begin{pmatrix} p(b) \\ u_2(b) \end{pmatrix}, \quad (1.22)$$

where k_{eq} is the bulk wavenumber of the wave propagating in the equivalent fluid and Z_{eq} is the medium impedances. The extension to PEM makes of a state vector that includes the fields, $(\hat{\sigma}_{12} \ u_2^s \ u_2^t \ \hat{\sigma}_{22} \ p \ u_1^s)^T$. More details are given in Chapter 11 of Allard and Atalla textbook².

For stacks of layers, an amount of N layers of the same nature are easy to manipulate since $T_{\text{global}} = \prod_{n=0}^N T_n$ since they have the same size. multiphysics with interface matrix that relate the relation between the various fields at an interface (non square matrices). This method was deep dived in several textbooks among which we propose Refs.^{2,14} for additional details. The TMM also allows for the computation of complex system by an assembly of elements including admittance matrices in order to consider Helmholtz Resonator (HR) embedded in an equivalent fluid layer¹⁵.

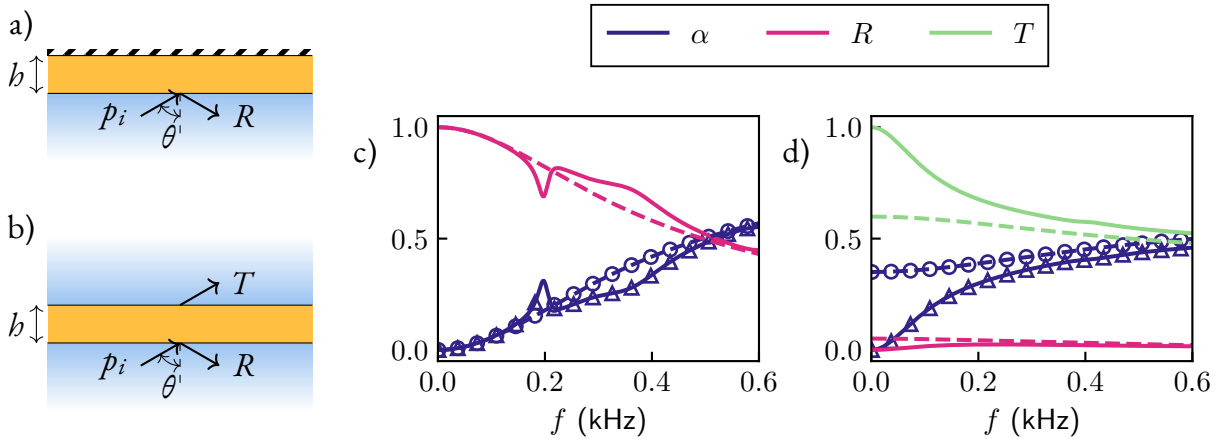


Fig. 1.3: Diagram for the a) reflection, b) transmission problem. Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the c) reflection, d) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid ($\circ$) and for the absorption coefficient (Δ)

While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes. From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients by simply manipulating the expressions for the fields contained in the state vector with the scattering coefficients.

Considering a rigidly-backed layer, and thus $u_2(b) = 0$, we obtain a reflection coefficient,

$$R_b = \frac{T_{11} - Z_{eq}T_{21}}{T_{11} + Z_{eq}T_{21}}. \quad (1.23)$$

In the case of the symmetric layer, the reflection and transmission coefficients are determined from the scattering matrix as well,

$$T_s = \frac{e^{-ikL}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}, \quad R_s = \frac{T_{12}/Z_{eq} - Z_{eq}T_{21}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}. \quad (1.24)$$

which is only valid for reciprocal, time-invariant and linear assumptions. The absorption coefficient for the same two configurations are calculated as $\alpha_b = 1 - |R_b^2|$ and $\alpha_s = 1 - |R_s^2| - |T_s^2|$ and are shown in Figs 1.3c and d with markers. It can be seen that the same results (up to the respective numerical precision of each method) are obtained.

1.3 | Guided waves for dispersion problems

In this section, we take an interest in similar configurations as previously, removing the excitation or forcing term in the equations. Solving the appropriate set of equations for a given set of parameters leads to the modes of the free system, that is the solution of the characteristic equation leading a specific behaviour of the wave propagation in the structure for the said parameters. The form of the modes of wave propagation in layer is discussed and various methods for their calculation of are presented.

The existence of an interface in the geometry leads to the propagation of not only the bulk waves but also of so-called surface waves¹⁶ where the interface itself acts as the support for propagation. Different types of waves propagate along the interface. An example can be water waves that follow the interface between air and water; most of the energy is localized at the interface. A similar behaviour exists for seismic waves where the interface of a solid substrate supports the propagation of Rayleigh (longitudinal motion) and Love waves (transverse motion) which both show an energy strongly decaying with depth. Scholte waves are waves that propagate along the interface between a fluid and a solid medium. In fluid layers, the propagating waves are similar to studying a shear horizontal (SH) wave in an elastic layer. One mode propagates at zero frequency, while countless other modes propagate with a cut-off frequency that corresponds to the successive quarter-wavelength resonances, essentially depending on the layer thickness and the nature of the boundary/interface conditions.

From the previously obtained linear systems in Section 1.2.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, in the general form $\mathcal{L}(\mathbf{k}, \omega)\mathbf{s} = 0$, where the operator $\mathcal{L}$ governs wave propagation in the structure, $\mathbf{k} = (k_1, k_2)^T$ is the wavenumber vector in the 2D geometry and $\mathbf{s}$ a vector containing the amplitude of all the partial waves. In order to obtain nontrivial solutions for $\mathbf{s}$, the system should satisfy a vanishing determinant for the linear system, $\det(\mathcal{L}(\mathbf{k}, \omega)) = 0$. The parameters satisfying this so-called *dispersion equation* can be sought, allowing to identify the modes of the structure. These equations can be solved for various parameter spaces depending on the perspective adopted on the problem^{17,18}: solving for frequencies while imposing the wavenumber or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve complex wavenumbers. The solutions are thus able to indicate on rather the spatially induced dispersion occurring in the system.

1.3.1. A bestiary of the wave solutions in dissipative and open waveguides

In this section, an overview of the possible sets of solutions for the guided waves in waveguide is presented. These sets of solutions are determined by the direction of the propagation, the attenuation

and the nature of the leaky wave if it exists. This *bestiary* is split into 3 categories. For all cases, the focus is on the values taken by the real (denoted by ') and imaginary parts (denoted by '') of each wavenumber component. Considering a layer and half-space system, there exists three wavenumber components from Eq. (1.16). The form of solutions are assumed as

$$p_l \propto e^{-ik'_1 x_1} e^{+k''_1 x_1} e^{\pm ik'_{2,eq} x_2} e^{\mp k''_{2,eq} x_2}, \quad p_0 \propto e^{-ik'_1 x_1} e^{+k''_1 x_1} e^{+ik'_{2,eq} x_2} e^{k''_{2,eq} x_2}. \quad (1.25)$$

The longitudinal wavenumber is the same in both layer and half-space domain as a consequence of the Snell-Descartes law and the continuity equations: there exists in total three dependant wavenumber components in this system, k_1 , $k_{2,0}$ and $k_{2,eq}$ among which determining two allows to compute all,

$$k_{2,0} = \pm \sqrt{k_0^2 - k_1^2}, \quad k_{2,eq} = \pm \sqrt{k_{eq}^2 - k_1^2}. \quad (1.26)$$

Because of the reciprocal nature of the presented configurations, and the considerations of unbounded layers in the x_1 direction, a plane of solutions corresponding to $k'_1 \geq 0$ can be selected, as the right-going solution is perfectly equivalent solutions to the left-going solutions.

In **conservative and closed systems**, the study is restricted to only the two wavenumber components in the layer, since no half-space is accounted. The total acoustic energy is thus confined to the structure enclosed by its boundaries. A type of wave solution exists as *perfectly guided*, this propagating wave has $k'_1 > 0$ with no attenuation $k''_1 = 0$. Solutions include *purely evanescent* wave as well $k''_1 > 0$ which often exists and continues a propagating mode below its cut-off frequency.

Open systems supports the existence of the previous types of modes with additional phenomena. Since part of the energy is susceptible to radiate out from the plate, the energy is no longer confined to the layer. Some of the previously purely guided modes can thus become attenuated, $k''_1 > 0$. Waves are still *perfectly guided* with conditions $k'_{20} = 0$ with evanescent waves in the fluid $k''_{20} < 0$, meaning that waves are confined to the proximity of the layer interface. The additional condition $k'_1 = 0$ correspond to *trapped modes*, which are a set of discrete eigenvalues of the system. They can exist in waveguides where the system is confined in one direction at least, the energy can thus be localized around some obstacle or defect by the evanescent waves^{19,20}. In open systems, they correspond to waves that are evanescent at infinity. However, in open waveguides, the acoustic energy is usually able to radiate outside of the waveguide, leading to two type of waves as *leaky waves* in the case $k'_{20} > 0$ and *incoming waves* (also called backward leaky) with $k'_{20} < 0$. These solutions are respectively exponentially growing as it radiates $k''_{20} > 0$ or attenuated $k''_{20} < 0$. These at first counterintuitive results ensue from the requirements that the wave should be propagating $k'_1 > 0$ and attenuated $k''_1 < 0$ along the layer, and also from the relation, $k'_1 k''_1 = -k'_{20} k''_{20}$ obtained by taking the imaginary part of the dispersion equation²¹. We refer also to the thesis of Kiefer²², especially Section 3.2 dedicated to the form of solutions for quasi-guided waves in the case of an elastic layer coupled to a semi-infinite fluid half-space.

The addition of **dissipation in open systems** leads to the mixing of several loss mechanisms: the leakage of waves and the dissipations within the guide (which can be of whatever nature). In this case, the separation between incoming and leaky waves is not possible anymore. This particular aspect has been very scarcely tackled in the literature: an unpublished paper is interested in an elastic waveguide²³ while in optics, a paper discusses the waves guided in nanospheres with a complex permittivity²⁴. There, the modes are sorted as *proper* and *improper* solutions. This complex case is covered in a dedicated section ?? in chapter ??.

Fig. 1.4: Classes of solutions for the different types of propagating waves in a-c) closed nondissipative guides. d-f) open non dissipative guides. f-h) open and dissipative guides.

In the next parts of this chapter, a discussion on available methods to compute the complex dispersion relation for guided waves in single or stacks of homogeneous layers is proposed. There are

semi-analytical methods that can be used, Section 1.3.2 or numerical methods that involve some sort of discretization in Section 1.3.3.

1.3.2. Computing the modes of porous layers using semi-analytical tools

The literature on the dispersion relation of porous/poroelastic-based structures is scarce; Groby *et al.*²⁵ have shown the complex dispersion relation in a homogeneous equivalent fluid layer coupled to air while Boeckx *et al.*²⁶ have computed the modes propagating in a poroelastic waveguide coupled at each top and bottom interface with an outgoing plane wave propagating in an air medium. This has been done using an analytical description of the system by expressing the interface conditions in a linear system (which we call GM in this work). Both of these work have used a root-finding procedure in order to solve the modes of the structure.

Once the linear system is obtained, we are faced with either or two of these possibilities. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible.

Using Eq. (1.19), root finding such that $\det(\mathcal{M}(k_1, \omega)) = 0$; the determinant of the matrix from Eq. (1.18) can be expressed as

$$2 \frac{k_{2,eq} k_{2,0}}{\rho_{eq} \rho_0} \cos(k_{2,eq} h) + i \left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2} \right) \sin(k_{2,eq} h) = 0. \quad (1.27)$$

From this expression, one can use a root-finding procedure to solve for either the term $k_{2,eq}$ or k_1 to obtain the dispersion relation by varying the frequency. Results are shown in fig. 1.5

Fig. 1.5: Dispersion relation of a rigidly-backed porous layer.

We reference to Appendix ?? for extensive details on the root-finding procedure based on the Müller method. These results show an infinite amount of modes [comment the fig.](#)

Generally, this kind of characteristic equation can only be obtained for very simple cases as is the case here, for a single fluid layer. The characteristic equation for Lamb modes is also well known²⁷. However, more complicated geometry requires to solve the root-finding iterative process by using directly a numerically-computed matrix determinant, or in many case, it is outright not possible due to the analytical complexity of the description of the system. In this case, more complex numerical method are thus required; a discretization of the geometry and an interpolation of the physical fields is done.

1.3.3. Numerics for modes computation

Regarding the computation of complex modes in homogeneous layer, several numerical methods have been investigated in the last decades. Guided waves represents a large topic and we focus here on the case of simple geometries; similar configurations are considered, an unbounded single (or stack of) layers with two infinite interfaces as presented in the section where we were interested in their acoustic response.

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM)²⁸, which has been especially used for the computation of guided wave modes in elastic layers^{29–31}. We refer to the introduction of Chapter ?? for an extensive bibliography on SCM applications.

This method is useful for the discretization of 1D geometries as is the object of the present section.

In the case of simple 2D geometries, a 1D discretization along the transverse direction associated to a harmonic motion in the longitudinal direction. Even if the use of finite elements seems farfetched,

this method could be relevant in contexts where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include [lister des publis ici](#). Examples are also given in Chapter ?? . A detailed description of the calculation of Lamb modes using a simple SAFE implementation using high-order Lobatto polynomials is proposed in Appendix. ?? . There, its performance is also compared to that of SCM.

- scm²⁹, sort of pml in scm³¹ vs local coupling³⁰
- safe has been widely used for this type of problems as well [cite paper with leaky waves and elastic layer + paper SAFE-PML 2.5D finite differences](#)¹⁷

There is a discussion to be made on the relevance of the imposition of a local coupling contrary to a full modelling of the surrounding media with some termination condition satisfying the Sommerfeld condition, PML or analogous absorbing layer. While leading to no assumptions on the radiating field nature contrary to imposing a local coupling, the modeling of the half-space by using some artificial absorbing layer at the end leads to the apparition of modes of the cavities³².

We argue that the advantages of the imposition of a local coupling is two-fold: less degrees of freedom are necessary since the sole interface is modeled, not the whole half-space. For dispersion problems, this is particularly interesting since we directly obtain leaky mode solutions, without needing to filter out all radiation modes linked to the additional cavity and the resonances of the PML

[relire, réécrire et compléter cette section](#)

gm, tmm, scm, safe, and fem

Fig. 1.6: Discretization for each method

décrire open plate model vs full plate fluid model

1.4 Methods for modelling metaporous surfaces and interfaces

1.4.1. A word on periodic structures

A large literature that shows the enhancement of absorption properties by periodicizing porous layers with scatterers was discussed in the introduction. In this section, a presentation of the fundamental notions surrounding periodic structures is proposed. In wave physics, periodic structures have been shown to introduce interesting effects towards applications for wave control and constitute the cornerstone of acoustic metamaterials. The periodicity can be imposed in acoustic systems using the so-called Bloch-Floquet theorem. This formalism has been pioneered by Felix Bloch and Léon Brillouin among others^{Bloch1929, 33}. Throughout this manuscript, we consider a periodic cell along only the longitudinal direction, that 1D periodicity which we denote d . Taking an infinitely repeating lattice or a crystal, the description of the wave propagation in only one of the so-called unit cell, one spatial period of the total structure, is enough to describe the whole structure. The layer of infinite extent is thus limited to a single unit-cell. A wavefunction $\psi(x)$ describing the propagation should

thus depend on a field with the same spatial periodicity as the lattice. To do so, the wavefunction is projected onto an infinite modal basis, that is Bloch modes,

$$\psi_q(\mathbf{x}) = e^{ikx_1}. \quad (1.28)$$

The wavefunction at the left boundary is thus related to that at the right boundary modulated by the spatial periodicity and some translation vector, that is $\psi(x + nd) = \psi(x)e^{ikd}$.

The possibility of reduction of the spatial considerations of the problem also leads to a reduction of the reciprocal wavenumber space. In periodic structure, we often restrict this reciprocal space to the first Brillouin zone, that is to wavenumbers restricted to $k_1 \in [0, \pi/d]$ when looking at the band structure. The consequences of periodic media are directly observable in the band structure: bands are usually folded when the wavenumber reaches this limit of the reciprocal space. This folding occurs at the Bragg frequency corresponds to a relation between the wavelength λ of the excitation signal and the periodicity of the crystal lattices, $d \sin(\theta) = n\lambda$, with θ the angle of incidence. This condition corresponds to destructive interferences: this phenomenon is a cause of the appearance of the famous Bragg bandgaps; in these bands of frequency, waves are unable to propagate. Variations of these phenomenon have been investigated in acoustic metamaterials: e.g. the addition of periodic resonators leads to hybridization stopbands.

1.4.2. Scattering of waves by periodic structures: overview of methods

Analytical methods for the evaluation of the wave propagation in periodic structures often require tedious derivations, as the frequency regimes of interest is usually high enough to require the description of the multiple scattering happening in the structure. The scatterer of the unit cell onto which the incident wave impinges acts as a secondary source, which scattered field is successively scattered by each neighboring obstacles and so on. Multiple Scattering Theory is a theoretical framework that has been developed to account for these phenomena and was used to compute the acoustic response of metaporous structures with rigid inclusions³⁴ and metaporoelastic surfaces with solid and shell inclusions¹⁰. Both configurations have been tuned to show a perfect absorption peak in the frequency spectrum.

The remaining papers discussing how incident pressure fields are affected and scattered by metaporous structures use FEM, for the most part through the interface of Comsol Multiphysics. This very versatile and powerful method has been used for decades in all fields of physics and thus won't be detailed in this thesis. It is recommended to the reader as an introduction on the basis of FEM, the textbooks from Atalla and Sgard³⁵ related to acoustics and vibration problems and that of Dhatt *et al.* for higher-order finite elements problems³⁶. This method allows for the discretization of the geometry via a mesh in order to solve the appropriate motion equations. This has allowed to use rather complex geometries, e.g spirals³⁷ and labyrinthine paths³⁸ as scatterers embedded in porous layers as discussed in the state of the art. Additionally, Gaborit *et al.*³⁹ have developed a method in order to couple FEM with Bloch modes expansion of analytical fields described by TMM. They have used it to compute the scattering coefficients of structure that include a periodic cell sandwiched by homogeneous layers with an embedded inclusion.

1.5 | Dispersion relations and band structure in periodic lattices

1.5.1. Methods for closed systems

One of the most popular methods for the computation of band structures in periodic systems made of classical lossless materials are the Plane Wave Expansion (PWE) which is semi-analytical. It relies on a Fourier transform of the motion equations, which allows an expression of the fields expanded onto Bloch modes, truncated for numerical applications. This method, while easy to implement presents important limitations due to the convergence of Fourier series and the matching of constituent materials for the lattice: materials should preferably be of the same nature for the calculation to go well¹⁴. Generalized as the Extended Plane Wave Expansion (EPWE), the method also allows to obtain the evanescent Bloch modes in Ref.⁴⁰, allowing for complex dispersion diagram investigation and to account for evanescent modes in finite 2D periodic lattices.

The complexity of the system prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods. As the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation. **summary of all thesis of magliacano with method (soft solids)** The WFE as well as FEM has also been widely applied to these geometries. Their modularity and versatility lead them to be good candidates for the modeling of our structure.

1.5.2. Unpromising adaptation of existing methods

The complexity of the system of interest in this thesis as presented in Section?? of the Introduction prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods: as the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

Multiple scattering could also be considered. However, it has been successfully used to describe the acoustic scattering of periodically arranged rigid inclusions in equivalent fluid unit cells³⁴ as well as solid inclusions in a poroelastic unit cell¹⁰. However, the so-called Schlömilch series that appear in the calculation, from the effect of the scattering of the neighboring cylindrical inclusions appear to be not so manipulable when considering complex longitudinal wavenumbers. Its expression is

$$\sigma_q^X = \sum_{l=1}^{\infty} H_q^{(1)}(k^X l d) \left((-1)^q e^{i k_1^X l d} + e^{-i k_1^X l d} \right), \quad (1.29)$$

and indeed, separating the k_1^X real and imaginary inevitably leads to at least one growing exponential term in the series, leading to it being unable to converge, up to our study.

We face challenges when trying to model this structure using the tools classically used in the literature, that is COMSOL Multiphysics. Although there are built-in tools in the software for "Eigenfrequency" studies, the library aims at solving complex frequencies of the system, given a set of parameters for the wavenumber. As it has been discussed earlier in the chapter, the goal of this work is to solve for complex wavenumbers and real frequencies. Even trying a manual implementation of the model in the "Weak Form PDE" interface, the tendency of this software to function as a black box introduces a new degree of complexity to this strategy. While FEM remains the soundest strategy for the modeling of the complex inhomogeneous geometry of the unit cell of metaporous surface, we propose to adapt in-house existing frameworks in order to develop numerical tools in this thesis.

1.6 | Prospects of methods for the thesis

1.6.1. Scattering properties of metaporous

We have seen that FEM seems to provide promising possibilities for the modelling of complex periodic cores. However, our considerations of open systems leads to question regarding the modeling of the half-spaces surrounding the structure.

1.6.2. Complex modes in metaporous

- Stacks of homogeneous layers described with a 1D discretization and study guided waves
- limitations of MST lead to the investigation of something else, as will be discussed in chapter ??

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